

Towards the ‘German chronometer’. The introduction of precision timekeeping in the German mercantile marine and Imperial Navy in the nineteenth century

Günther Oestmann*

Soon after the formation of the German Empire in 1871 ambitious efforts started to build up a Navy, and the Imperial Admiralty called for a self-sustaining manufacture of nautical instruments and chronometers. In 1877 annual competition trials for testing chronometers commenced at the German Hydrographical Institute (Deutsche Seewarte), and in 1910 the target of creating the ‘German chronometer’ had been met. But this was a narrowly defined success, in that the import of English chronometers continued. The production of chronometers in Germany was limited by the nature of demand, inherent supply problems and partly encouraged, but also constrained, by imposed governmental regulation.

The introduction of precision timekeeping in the German mercantile marine and Imperial Navy is an interesting case study for the establishment of a highly demanding and specialized manufacture between governmental regulation and technical limitations.

It has often been assumed that the introduction of chronometers was initially hindered by a certain conservative attitude among seamen and especially by their high cost. According to contemporary travel literature there is indeed evidence that there were few chronometers on German ships in the first half of the nineteenth century,¹ but

a differentiated analysis of the extant sources reveals that the spread of this new aid to navigation – the ‘soul of modern navigational calculation’, as Wilhelm von Freeden (1822–1894), director of the navigation school of Elsfleth put it² – depended on the direction of maritime trade.

In the beginning of the nineteenth century Prussia was not a major maritime power. Nevertheless chronometers were used in navigational training and practice, albeit under special conditions. The ships of the Prussian Mercantile Company (*Preußische Seehandlung*), founded in 1772, were very well equipped with modern

**Dr Günther Oestmann (oestmann@nord-com.net) is a lecturer at the Technical University of Berlin, Institute for Philosophy, Literature, History of Science and Technology. This article is an annotated version of a lecture given to the AHS at Greenwich on 17 May 2014. It summarizes the results of a research project funded by the German Research Council (Deutsche Forschungsgemeinschaft). In AH March 2013 a review was published of the author’s monograph on the subject: Auf dem Weg zum „Deutschen Chronometer“: Die Einführung von Präzisionszeitmessern bei der deutschen Handels- und Kriegsmarine bis zum Ersten Weltkrieg [= Deutsche Maritime Studien, 21], Bremerhaven 2012.*

1. E. g. Gottfried Duden, *Bericht über eine Reise nach den westlichen Staaten Nordamerika’s und einen mehrjährigen Aufenthalt am Missouri (in den Jahren 1824, 25, 26 und 1827), in Bezug auf Auswanderung und Ueberbevölkerung [...]* (Elberfeld: Samuel Lucas, 1829), p. 11; Fridrich Arends, *Schilderung des Mississippithales [...], oder des Westen der Vereinigten Staaten von Amerika: Nebst meiner Reise dahin* (Emden, H. Woortman: 1838, repr. Leer, Schuster: 1974), p. 17.

2. Wilhelm Ihno Adolph von Freeden, *Handbuch der Nautik und ihrer Hilfswissenschaften* (Oldenburg, Schulztesche Buchhandlung: 1864), preface, p. IV.

navigational instruments, including chronometers.³ This was a heavily subsidized state company however, and therefore an exceptional case. The introduction of the chronometer in Prussia was more the result of all-embracing governmental regulation than arising from actual nautical requirements. Prussia only had a number of small harbours on the shores of the Baltic Sea and the port of Emden. Maritime trade was mainly limited to adjacent countries (especially England and Scandinavia), ships travelled alongside shorelines and seldom crossed the open sea. As local knowledge of navigable waters was sufficient there was no need for the determination of geographical coordinates. But the Prussian government established a highly ambitious program for nautical education and tried introducing new methods and instruments, sometimes much to the resentment of 'old salts'. In 1823, 1824, 1840 and 1862 comprehensive instructions for teaching and examination were issued with ever more stringent requirements. Prussia was a latecomer on the ocean, and this fact may explain the ambitious endeavours in maritime affairs and the attempt to do everything with the utmost correctness. Around 1850 the entire tonnage of the Prussian mercantile marine was only two thirds of that of Bremen, and Prussian ships still mainly navigated on the North and Baltic Sea. But in 1858/59 there were 192 chronometers on 183 ships,⁴ and approximatively one quarter of Prussian

ships had a chronometer on board.

In the Kingdom of Hanover shipping traffic also was regional with only few exceptions, and many small ships from East Frisia carried neither a sextant, nor a chronometer. In 1865 906 ships were registered, but only 16 per cent (145 ships) possessed a chronometer.⁵ Of the 188 ships registered in Papenburg thirty-eight possessed a chronometer in 1867.⁶

The situation in major ports (e.g., the Free Hanseatic City of Bremen) with overseas trade was substantially different. Merchants and shipowners there were early 'global players'. They sent their ships all around the globe, and the application of chronometers was a fairly rapid development: Already in 1842 140 out of 215 ships registered in Bremen had a chronometer, and in 1848 this had risen to 202 out of 246.⁷ In 1876 every ship was equipped with at least one chronometer.

The vast majority of chronometers were imported from England. Until the mid-nineteenth century only few chronometer makers were active in Germany. Johann Georg Thiele of Bremen (1714–1784) built the first chronometer in Germany of highly original construction almost immediately after Harrison, but his timekeeper failed when Nevil Maskelyne put it on trial at Greenwich in 1778.⁸

Early pioneers of precision timekeeping in German territories were Christian Friedrich Tiede (1794–1877),⁹ Friedrich Adolph Nobert (1806–1881) and Johann

3. Heinrich Berghaus (ed.), *Sechs Reisen um die Erde der königlich preussischen Seehandlungs-Schiffe Mentor und Prinzeß Louise innerhalb der Jahre 1822–1842: Auszug aus den Schiffs-Journalen in Bezug auf Hydrographie und Meteorologie* (Breslau, Grass, Bath & Comp.: 1842), pp. 1–7; Johann Friedrich Meuss, *Die Unternehmungen des Königlichen Seehandlungs-Instituts zur Emporbringung des preussischen Handels zur See: Ein Beitrag zur Geschichte der Seehandlung (Preussische Staatsbank) und des Seewesens in der ersten Hälfte des neunzehnten Jahrhunderts*, Veröffentlichungen des Instituts für Meereskunde an der Universität Berlin, n.s., vol. 2 (Berlin, Ernst Siegfried Mittler & Sohn, 1913), pp. 205–216.

4. Berlin, Geh. Staatsarchiv PK: I. HA Rep. 120 MfHuG C XVII 6a, Nr. 9.

5. *Die Rhederei Hannovers: Alphabetisches Verzeichniss aller Seeschiffe des Königreichs Hannover*, 6, (1866).

6. Jürgen Meyer, *Vom Moor zum Meer: Papenburger Schifffahrt in 3 Jahrhunderten* (Papenburg, Fehn-Verlag Eissing: 19892), p. 255.

7. Bremen, State Archive: 2-R.11.e.1 (*Schiffslisten und statistische Materialien über die unter bremischer Flagge fahrenden Schiffe*, 1765–1876).

8. For the details see Oestmann, *Auf dem Weg zum „Deutschen Chronometer“*, pp. 15–21.

9. [August Engelbrecht], 'Friedrich Tiede', *Allgemeines Journal der Uhrmacherkunst*, 16 (1891), 260–261; Wolf-Dieter Finck, 'Christian Friedrich Tiede: Königl. Astronomischer und Hof-Uhrmacher in Berlin', *Alte Uhren*, 5 (1982), H. 1, 9–20.

Christian Hanneke (1778–1848). Heinrich Johann Kessels (1781–1849) worked in Altona, at the time belonging to Denmark. Although he relied on the import of movements from England – or to be more specific, from his business partner George Muston in Bristol – the aftermath of his workshop was very important for the establishment of chronometer manufacturing in Altona and Hamburg.¹⁰ Jacob Auch (1765–1842), Johann Christian Friedrich Gutkaes (1785–1845) and Johann Heinrich Seyffert (1751–1818) produced chronometers for astronomical and surveying purposes only.

In 1867 the North German Federation (*Norddeutscher Bund*) was founded, and efforts were made towards unification of navigational teaching. Regulations were basically adjusted to the comparatively high Prussian standards, because Prussia dominated the negotiations and held the majority in the Federal Assembly.

The institutional framework as well as the development prospects of the Prussian Navy changed fundamentally after the proclamation of the German Empire. Albrecht von Stosch (1818–1895), a general of the infantry, became the first head of the newly founded Imperial Admiralty in 1872. With a continuous reliance on foreign imports (particularly from England) he anticipated difficulties in replenishment in future combats. Therefore von Stosch tried promoting domestic industry, called for a self-sustaining manufacture of nautical instruments and chronometers and launched national examination standards. In conjunction with these ambitious efforts and the growing demand, a new industry of precision mechanics gradually developed in Germany.¹¹

Already in 1861 a ‘Hydrographic Bureau’ of the Prussian Admiralty had been

installed.¹² Apart from nautical cartography and researches on the problem of compass compensation (which became a very important issue after the introduction of iron shipbuilding) the procurement and equipment with instruments was an area of responsibility. The Imperial shipyards in Wilhelmshaven, Kiel and Danzig set up depots of instruments on their premises.

For nautical and scientific issues of the German mercantile marine the German Hydrographical Institute (*Deutsche Seewarte*) in Hamburg was responsible. This institution had emerged from a private enterprise established in 1867 by Wilhelm von Freeden, director of the navigation school in Elsflath, and was subjected to the Imperial Admiralty. In 1875 work started as a national institute. The geophysicist Georg Balthasar Neumayer (1826–1909), who had already been in charge as ‘Hydrographer of the Admiralty’, was appointed as its first director (Fig. 1). Neumayer was a skilful organizer of scientific work and developed a tremendous activity during his long term of service until 1903. By his researches in geophysics and his contributions to polar research he achieved international fame.

At first the *Seewarte* consisted of three departments: I. Maritime Meteorology and Oceanography, II. Procurement and examination of instruments, publishing, III. Meteorological telegraphic service and storm warnings.

George Rümker (1832–1900), director of the astronomical observatory in Hamburg and inspector for nautical examinations, advocated the establishment of a testing laboratory for chronometers. On behalf of the Hamburg authorities Rümker in 1876 visited the ‘Special Loan Collection of Scientific Instruments’ in the South Kensington Museum in London and took a closer look at the chronometer testing

10. On Kessels’ life and work see Günther Oestmann, *Heinrich Johann Kessels (1781–1849): Ein bedeutender Verfertiger von Chronometern und Präzisionspendeluhren, Biographische Skizze und Werkverzeichnis*, Acta Historica Astronomiae, 44 (Frankfurt/M., Harri Deutsch, 2011).

11. The authoritative study on Stosch is the PhD thesis of Dirk Sieg: *Die Ära Stosch: Die Marine im Spannungsfeld der deutschen Politik 1872 bis 1883*, Kleine Schriftenreihe zur Militär- und Marinegeschichte, 11 (Bochum, Verlag Dr. Dieter Winkler: 2005); see especially pp. 391–396.

12. On the history of this institution see *50 Jahre vom Hydrographischen Bureau des Königlich Preußischen Marine-Ministeriums zum Nautischen Departement des Reichs-Marine-Amtes 1861–1911* (Berlin, Ernst Siegfried Mittler & Sohn: 1911).



Fig. 1. Georg Balthasar Neumayer, first director of the German Hydrographical Institute (*Deutsche Seewarte*) in Hamburg. (Hamburg, Archive of the Federal Maritime and Hydrographic Agency)

facilities in Greenwich and Liverpool (Bidston in Birkenhead). The experiences gained there were to be exploited for the establishment of a similar institution at Hamburg, which became the fourth department of the *Seewarte*.

Already in 1877 annual competition trials were arranged for testing chronometers in accordance with the common procedures of the Royal Observatory at Greenwich and Liverpool Observatory.¹³ The facilities were at first located on the premises of Hamburg observatory in a separate building with an ice cellar and a gas-heated oven especially constructed for testing chronometers (Figs 3 and 4). In 1892 the department moved into a new building erected in the immediate vicinity of the *Seewarte* with improved and considerably enlarged facilities (Fig. 2).

Two other institutions for chronometer trials were created. Since 1873 chronometers had been tested in Kiel Observatory, albeit



Fig. 2. The new chronometer testing station of the *Deutsche Seewarte* in Hamburg, put into service in 1892. (Hamburg, Archive of the Federal Maritime and Hydrographic Agency)

under unsatisfying conditions. Therefore the Imperial Admiralty planned an observatory especially designed for the purpose of examining and testing chronometers, which was completed in 1883 and situated near the Naval Academy. Two boxes heated with warm water were available, but there was no ice cellar. The observatory also operated a time ball on the Imperial Shipyard. When the ball dropped at 12 o'clock Kiel meantime, a cannon shot was fired from a look-out ship in the harbour (Fig. 5). This moment was a reminder for lunch time and called 'soup shot' in common parlance.

In Wilhelmshaven a naval observatory was built in 1874–78, but in contrast to the chronometer observatory in Kiel, which served one purpose only, the tasks of this institution were conceived in a broader way. Apart from chronometer trials and time service, geomagnetic and meteorological observations and researches

13. For a summary treatment see Kuno Heuer, *Rückblick auf die in den Jahren 1877 bis 1917 auf der Deutschen Seewarte abgehaltenen Chronometer-Wettbewerbs-Prüfungen*, Aus dem Archiv der Deutschen Seewarte, 38, nr. 3 (Hamburg 1920).

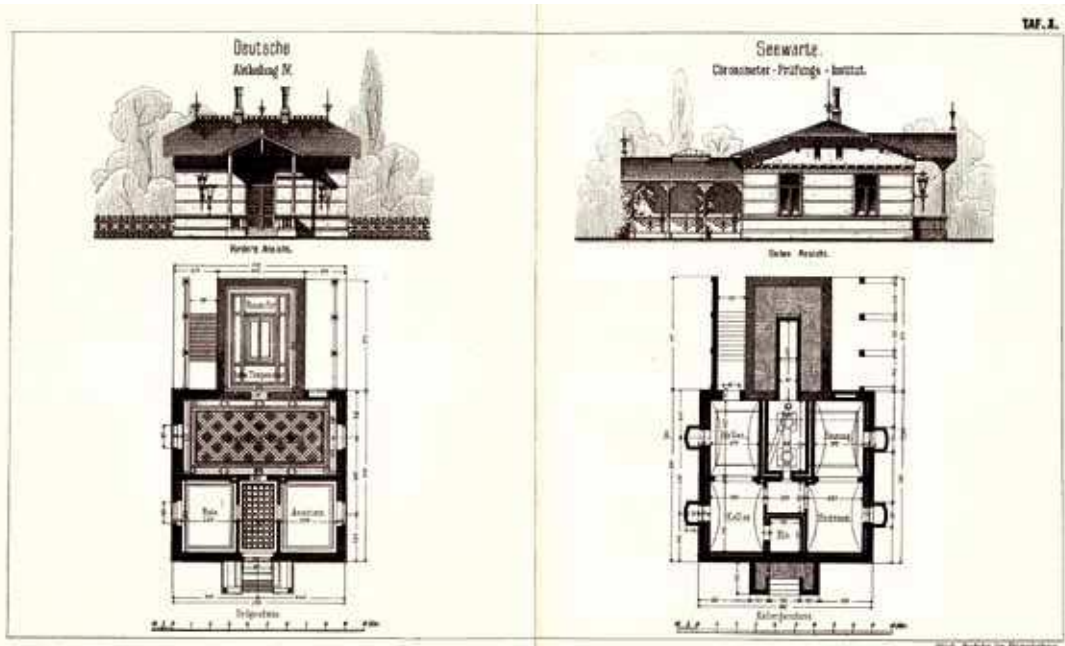


Fig. 3. The first building for testing of chronometers erected in 1876 on the premises of the Hamburg astronomical observatory, with ice cellar. (Eugen Gelcich, *Geschichte der Uhrmacherkunst*, Weimar 1892⁵, plate X).

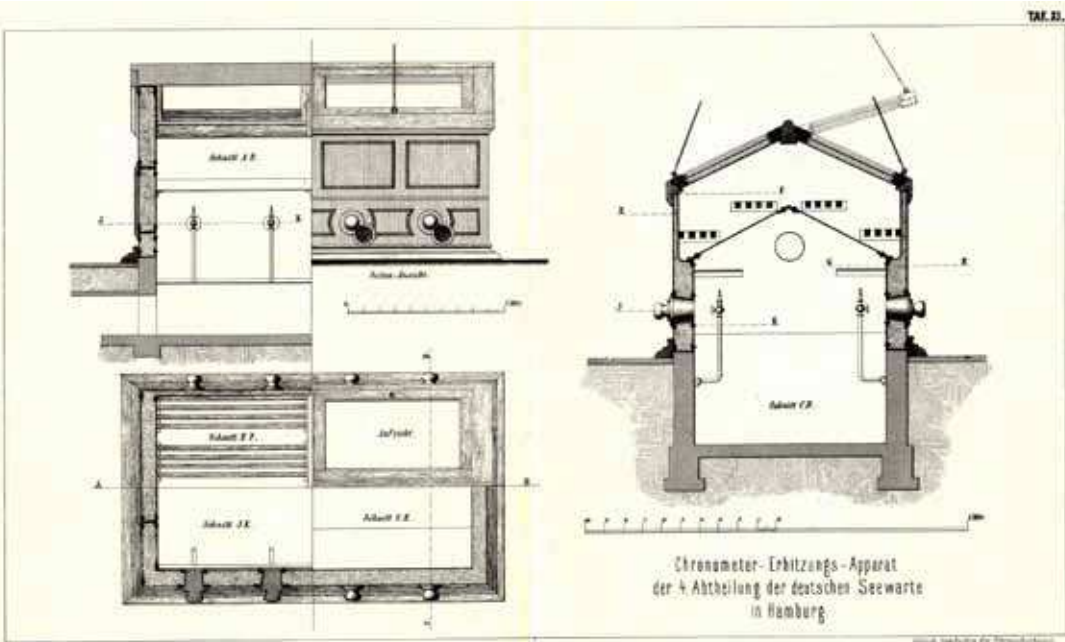


Fig. 4. The gas-heated oven in the first building for testing of chronometers. (Eugen Gelcich, *Geschichte der Uhrmacherkunst*, Weimar 1892⁵, plate XII)

on tide prediction were carried out. The facilities for testing chronometers were very simple, however: neither an ice cellar nor a heating apparatus was available. The radiators of the central heating acted as a

makeshift for the latter, and tests in lower temperatures were conducted in winter by opening the windows. Sometimes the required temperature of 5° C could not be reached due to mild weather.



Fig. 5. The firing of the noon cannon in the harbour of Kiel was the signal for citizens to set their clocks and pocket watches. Top, left to right, the chronometer observatory outside and inside, and the electrically released timeball on the Imperial Shipyard. Wood engraving in *Leipziger Illustrierte Zeitung*, 1890 after a drawing by Ferdinand Lindner (1847–1906). (Kiel, Municipal and Maritime Museum)

Only very few original sources relating to both institutions are extant, because the buildings and almost all records have been destroyed in the Second World War.

In Kiel, Wilhelmshaven and Hamburg a variety of special studies on the behaviour of chronometers were undertaken, for example whether shocks and vibrations of the firing of guns caused malfunctions or how the influence of atmospheric pressure, humidity and transport conditions affected a chronometer's rate. In 1883 the *Seewarte* was equipped with a special rotating device driven by a gas motor in the cellar, which

had been originally designed for the examination of anemometers ('Combe's Apparatus', informally called 'merry-go-round' by the staff; Fig. 6). The machine could also be used for the examination of chronometers by adding a special device for the simulation of rolling and pitching of a ship.¹⁴

After the commencement of chronometer trials the first 'chronometer conference' was held in Hamburg in 1878, where test procedures and an authoritative mode for examinations were discussed.¹⁵ The second item on the agenda concerned the 'exclusive

14. The apparatus is first mentioned in Neumayer's description of the *Seewarte*: *Die Deutsche Seewarte: I. Beschreibung der Zentralstelle in Hamburg*, Aus dem Archiv der Deutschen Seewarte, 7, 1884, nr. 2, (Hamburg 1885), pp. 8, 13–15. It had been built 'according to instructions given by the director' by the mechanics W. Ritter in Altona and Frank von Liechtenstein in Hamburg. For a description see also 'Der Combesche Apparat in seiner Anwendung als Chronometer-Schaukelapparat', *Deutsche Uhrmacherzeitung*, 29 (1905), 120–121.

15. The minutes of the three 'chronometer conferences' held in 1878, 1887 and 1898 (Freiburg i. Br., Federal Archive/Military Archive) are printed in Oestmann, *Auf dem Weg zum „Deutschen Chronometer“*, pp. 395–425.



Fig. 6. 'Combe's Apparatus' in the atrium of the Seewarte, used for the examination of anemometers and chronometers. The device was driven by a gas motor in the cellar. (Hamburg, Archive of the Federal Maritime and Hydrographic Agency)

admission of German work' demanded by the *Seewarte*: 'It is notorious that chronometer manufacturers obtain their instruments generally from England and only put finishing touches to the adjustment.' In this matter the participants were rather pragmatic and stated that chronometer makers could not be called to produce all components themselves, but making specific parts should be required.

On the authority of the Imperial Admiralty a second conference took place in Hamburg in 1887. The question of admission criteria was on the agenda again – Swiss makers like Ulysse Nardin, Henri Grandjean and Henri-Robert Ekegrèn had participated in the trials, and some conference members pleaded for general liberalisation. This proposal was dismissed to protect German makers. Concerning the parts they had to produce themselves there was still consensus that it was not necessary to alter existing regulations. Only 'working over' (*Durcharbeitung*) of balance and spring and rating should be mandatory. From 1889/90 onwards chronometers handed in for trial were graded in four classes according to defined limits of error.

There was a good reason why the discussion of origin came up again and again. England with its two market leaders Kullberg¹⁶ and Mercer¹⁷ dominated fabrication and trade of chronometers almost entirely, and the majority of German makers (Knoblich, Ehrlich, Lidecke, Diederich, Nieberg, Petersen, Eppner and others) relied on complete chronometers, movements, blanks and spares from England. By a systematic survey of the company

archives of Kullberg and Mercer the extent of dependency could be specified in detail.¹⁸

In Kullberg's Manufacturing Books preserved in Guildhall Library (London) each chronometer is assigned a separate page. The heading contains the name of the customer and the date of delivery of the finished chronometer. At the beginning of the first column on the left the work number of Kullberg is noted, followed by an array of individual commissions for external craftsmen (Movement/ Escapement Jewelling/ Mainspring/ Escapement Wheel/ Pallets/ Balance/ Pivoting/ Planting/ Stud/ Brass Box/ Finishing/ Chain/ Jewelling/ Cutting/ Springing/ Timing/ Examining/ Hands/ Box/ Gymbles/ Engraving Dial/ Engraving Name/ Engraving Tablets/ Winder/ Outside Box/ Sundries [various expenses]) with delivery dates (in/out) and the incurred costs. Kullberg's principal provider of blank movements was Joseph Preston in Prescott.

Of particular interest is the category 'Engraving Dial', because here the 'transubstantiation' to a German, Dutch, Russian (or whatever) chronometer takes place, which is connected with a change to the work number of the purchaser.

The Order Books of Kullberg contain further information concerning orders of German makers and consist of chronologically ordered partial transcripts and summaries of incoming correspondence, often accompanied by serial numbers. Adolph August Kittel (1846–1921), who manufactured complete chronometers, was a notable exception.¹⁹ In a sense he embodied the antithesis to Knoblich and his successor Meier, who always relied on

16. On Kullberg see Alun C. Davies, 'The House of Kullberg', *AH*, 35 (2014), 635–648 and his excellent, unfortunately unpublished, PhD thesis *The Rise and Decline of the British Horological Industry c. 1620 to c. 1920*, Queen's University of Belfast 1986, pp. 199–210.

17. Tony Mercer, *Mercer Chronometers: Radical Tom Mercer and the House he founded* (Ashford, Brant Wright Associated Ltd.: 1978).

18. A compilation of German orders from Kullberg covering the years 1873–1914 can be found in Oestmann, *Auf dem Weg zum „Deutschen Chronometer"*, pp. 261–276; for Mercer's deliveries see pp. 284–296. The company archive of Mercer is located in the National Museums Liverpool and consists of 22 boxes containing work books, account books, payrolls, documents, pamphlets, photographs &c., but most of the material is more recent. The Manufacturing Books on which Tony Mercer's account is obviously based (*Mercer Chronometers*, pp. 151–236) have not been transferred to Liverpool, and their present whereabouts are unknown.

19. On Kittel's life and work see the entry in the *Biographisches Lexikon für Ostfriesland* by the author <<http://www.ostfriesischelandschaft.de/index.php?id=776>>



Fig. 7. Chronometer No. 269 by Kittel (Canberra, National Museum of Australia) with toothed barrel and pivoted detent, purchased in 1907 by the Carnegie Institution of Washington. (Photo Jason McCarthy, National Museum of Australia)

English imports. With enduring persistence – one may even say stubbornness – Kittel devoted himself to the creation of the ‘purely German chronometer’, and the problems he had to fight throw a revealing light on the particular circumstances and difficulties of the establishment of an independent German chronometer production.

Already in 1887 Leo von Caprivi (1831–1899), head of the Imperial Admiralty and successor of von Stosch, referred to Kittel and Dencker as makers ‘who are standing on their own feet’ but also labelled them ‘restless minds’.²⁰ This appreciation proved to be absolutely correct, because both Kittel and Dencker kept the naval authorities constantly busy over years.

Between 1878 and 1910 Kittel submitted seventy-five chronometers to the annual trials of which the Imperial Navy purchased forty-nine for 36,850 Reichsmarks. Moreover, he received fifteen awards totalling 11,800 Reichsmarks. Despite these obvious successes, Kittel’s reputation was not undisputed. The construction and execution

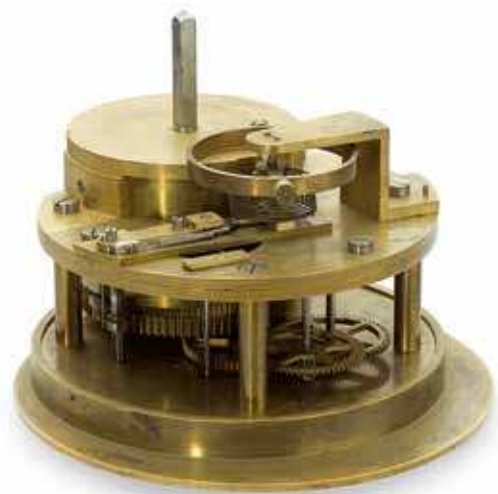


Fig. 8. Experimental chronometer movement with toothed barrel and constant force escapement, probably made by Adolph August Kittel. (Horological school, Hamburg; photo Egbert Laska, Deutsches Schiffahrtsmuseum)

of his chronometers was criticized repeatedly, and Carl Rottok (1854–1931), head of the chronometer observatory in Kiel, remarked in 1907: ‘Ship commanders are always reluctant to take chronometers by Kittel, as they find there is always something with these instruments.’²¹ Due to the numerous petitions and complaints he wrote to the German Imperial Naval Office (*Reichsmarineamt*) Kittel was regarded as a rather querulous person.

Obtaining German-made chronometer chains was impossible until 1906, and therefore Kittel constructed a calibre of his own with toothed barrel and differential gearing for the up-and-down indicator. However, the abandonment of the fusee also had its disadvantages and was controversial. Nevertheless Kittel often achieved excellent rates. He also experimented with several auxiliary compensations of his own design, which were manufactured in Glashütte by Carl Bernhard Kohl (1832–1908) and later Richard Griebßbach (1868–1948). Kittel referred to his constructions as ‘spoke compensation’, ‘additional compensation’ or

20. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10774, fol. 75r.

21. ‘Die Schiffskommandos sträubten sich stets dagegen, die Chronometer von A. Kittel anzunehmen, denn bei diesen Instrumenten sei immer etwas los’ (Freiburg i. Br., Federal Archive/Military Archive: RM 3/10781, fol. 165v).

‘retrograde acting auxiliary compensation’. The latter consisted of small, bimetallic strips fitted with additional weights, which were mounted on the two masses of the balance. Contrary to Kittel’s claim this device was no original invention however, and had in fact been designed by the French maker Simon Vissière (1822–1887). For the escapement Kittel chose a variant of the pivoted detent similar to a construction by the French chronometer-maker Jean François Henri Motel (1786–1859).

Kittel was a stubborn loner with quirky features. As a tireless, highly original experimenter he did not care much about economic issues and died in poverty shortly after the First World War.

His colleague Ferdinand Dencker (1837–1917), who had always rejected participating in chronometer trials, was a staunch advocate of rationalized machine production of chronometers on an industrial scale and a severe critic. In 1897 he stirred up a public debate on the ‘foreign origin’ of chronometers by launching articles in horological journals containing direct assaults on the *Seewarte* as well as on several of his fellow makers. This very year was a critical juncture, because Rear Admiral Alfred Tirpitz (1849–1930; Fig. 9) had just been appointed as Secretary of State of the Imperial Navy Department, and in 1898 he introduced the First Fleet Act for the reorganization of Germany’s sea power, soon to be followed by the Second Fleet Act for the build-up of the so-called ‘risk fleet’ (*Risikoflotte*), which initiated a maritime arms race between Germany and England in 1900. The ‘chronometer question’, hitherto a rather remote debate in specialist circles of the Imperial Navy, now became a highly political issue.

At first Neumayer hoped to weather the storm, but a third chronometer conference was summoned immediately and convened in 1898. Now regulations were intensified and enactments for creating the ‘German chronometer’ decreed. Only freelance makers residing in Germany were allowed to participate in the annual trials, and with the exception of very few special parts that



Fig. 9. Rear Admiral Alfred Tirpitz, whose Fleet Acts turned the ‘chronometer question’ into a highly political issue. (Bremerhaven, Deutsches Schiffahrtsmuseum)

could not be produced in adequate quality (mainsprings and fusee chains), everything had to be manufactured in Germany. Chronometers with ‘foreign’ parts were neither acquired by the Admiralty nor could they compete for prizes or were entitled to awards. Makers had to fill out a detailed questionnaire concerning the origin of every part,²² and an expert commission meticulously examined chronometers submitted for trial. After Neumayer’s retirement Rear Admiral Alfred Herz (1850–1936) was appointed as successor in 1903. Under his direction (which lasted until 1911) the ‘German chronometer’ materialized at last.

But the imposed restrictions inevitably led to shortages, and compromises had to be found. Moreover, a profound technological change in chronometer compensation took place shortly after the turn of the century. This was connected with a special problem, namely the removal of the so-called ‘secondary error’ in

22. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10781, fol. 80r–81r; RM 3/10782, fol. 98r–99r; RM 3/10784, fol. 39r/v.

compensation at medium temperatures,²³ already discovered by Ferdinand Berthoud (1727–1807) in 1771²⁴ and later described by Dent (1790–1853).²⁵

Many constructions of auxiliary compensations had been proposed and tested with more or less promising results,²⁶ but only few chronometer makers were able to adjust these mechanisms properly.

Charles Edouard Guillaume (1861–1938), a Swiss-born physicist who worked at the International Bureau of Weights and Measures at Sèvres in France, had studied the properties of nickel steel alloys with the objective of making temperature-insensitive and less costly prototype standard measures for the length of the meter. In 1897 he had created an iron-nickel alloy (later called Invar) whose expansion coefficient was practically zero over a large range of normal temperatures.²⁷ A slight quantitative modification of the alloying components enabled the construction of a highly effective compensation balance, of which the first types were made by Nardin and Ditisheim in 1899. Owing to its superb performance the nickel steel balance was quickly accepted and made auxiliary compensations redundant. From 1911 onwards all German

chronometers submitted for trial were equipped with this new type of balance. But Guillaume had granted a licence for its production to a company in Switzerland (Société des Fabriques de Spiraux Réunies S.A. in Neuchâtel), and consequently German makers had to rely on imports. The question arose whether chronometers with imported balances should be admitted for trial.²⁸ This was granted, and in 1907 Lange & Söhne obtained a licence for the import of nickel steel and exclusive production of chronometer balances,²⁹ which were produced under contract by Richard Griebach in Glashütte (Fig. 10). But he manufactured balances made of nickel steel produced in Germany for himself also, which led to ongoing disputes between him and Lange.³⁰ Because Griebach was the only manufacturer available in Germany, the Imperial Naval Office withdrew its planned enactment that from 1910 onwards only German balances should be allowed and conceded the use of foreign imports. The same went for palladium springs made in Switzerland (this non-magnetic material proved to be useful for chronometers used in submarines). These exemption clauses were retained during the entire First World War.

23. When the temperature rises, the elasticity of the spring decreases. Therefore its accelerating force upon the balance diminishes, and the timekeeper goes slower. The opposite effect occurs at lower temperatures. To compensate this effect a bimetallic balance with split rim and attached weights is used. When the temperature rises, the rims tend to bend inwards, thus diminishing the moment of inertia of the balance. The action of the balance is controlled by the difference in expansibility between the brass and the steel of its rim. This expansion is expressed by two functions in which the coefficients are very approximately the same. The algebraic sum of these will be linear. But the elasticity modulus of the balance spring varies at different temperatures according to a certain quadratic curve, and the algebraic sum of the three curves will leave a quadratic remainder. Therefore the rate of a chronometer is correct at two temperatures only and it will gain at intermediate temperatures.

24. Paul Ditisheim, 'Ferdinand Berthoud et l'erreur secondaire de la compensation des chronomètres', *Journal suisse d'horlogerie et de bijouterie*, 67 (1942), 289–297 (292).

25. Edward John Dent, 'On the Errors of Chronometers, and a New Construction of the Compensation Balance', *Nautical Magazine*, 11 (1842), 760–768.

26. Richard Yrk, 'Hilfs-Kompensationen', *Deutsche Uhrmacher-Zeitung*, 22 (1903), 222–224, 263–266, 281–283, 337–338, 354–355; Rupert T. Gould, *The Marine Chronometer: Its History and Development* (London, Holland Press: 1978⁶), pp. 179–200.

27. Charles Edouard Guillaume, 'Invar and Elinvar', *Nobel Lectures: Physics 1901–1921*, ed. Nobel Foundation (Amsterdam, Elsevier: 1967), pp. 444–473. <http://www.nobelprize.org/nobel_prizes/physics/laureates/1920/guillaume-lecture.html>

28. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10780, fol. 14r–45v, 230r–232r.

29. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10781, fol. 157r/v.

30. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10780, fol. 230r–232r; RM 3/10781, fol. 71r–74r; RM 3/10782, fol. 165r–225r.

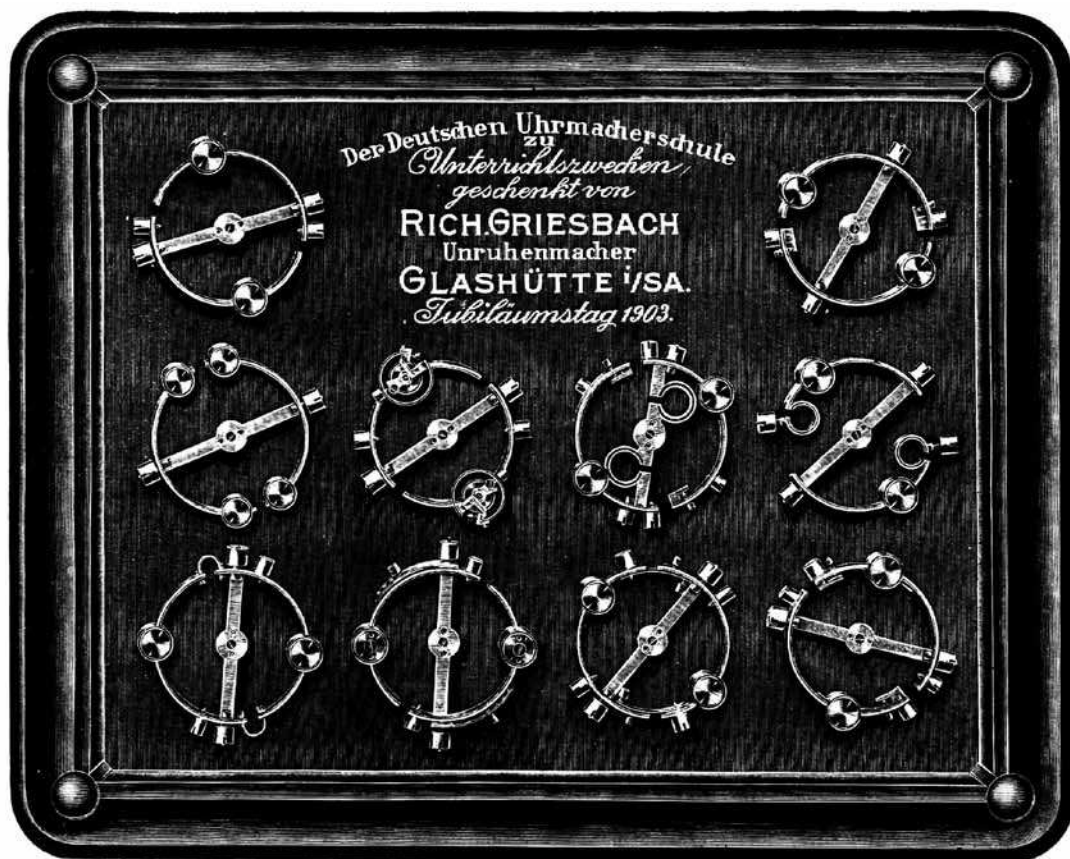


Fig. 10. Set of compensation balances, presented to the German watchmaking school in Glashütte for teaching purposes in 1903 by Richard Grießbach, balance maker at Glashütte. Formerly in the Horological Museum Glashütte; illustration from *Deutsche Uhrmacherzeitung*, 1904, 152. Above on the left: common bimetallic balance, on the right: Kullberg's first auxiliary compensation balance. Middle row (from left): Guillaume's nickel steel balance, balance with auxiliary compensation by Vissière/Kittel, Kittel's 'spoke compensation', balance by F. J. Garrard. Below (from left): Balance with 'rein compensation' (*Zügelkompensation*) by Johann Ernst Adolph Uhrig introduced in Germany by Johann L. Ehrlich, balance by John Poole, balance with spring auxiliary compensation by Kittel and Kullberg's second auxiliary compensation balance.

Since 1906 chronometer chains were manufactured by William Weicholdt (1866–1944) in Glashütte, and in the following year regulations concerning mainsprings and chains were further tightened.³¹ From 1908 onwards chronometers without fusees were no longer admitted,³² and Kittel, who had already completed a number of chronometers with toothed barrels for trial, was forced to look for alternative marketing

opportunities. He succeeded in selling several chronometers to the Carnegie Institution of Washington. (see Fig. 7)

As an astronomer Wilhelm Foerster (1832–1921) had a keen interest in time measurement. He was involved in the establishment of a system of electrically controlled public clocks in Berlin, as well as in the control of time-ball signals along German coasts.³³ In 1899 he founded the

31. *Jahresbericht über die Tätigkeit der Deutschen Seewarte*, 28 (1905), p. 39.

32. Freiburg i. Br., Federal Archive/Military Archive: RM 3/10782, fol. 48r.

33. Klaus-Harro Tiemann, 'Zur internationalen Verbreitung und Präzision von Zeitbällen 1866 bis 1887', *Uhren und Schmuck*, 26 (1989), 148–151, 156; 'Die 1880 unterbreiteten Vorschläge des Berliner Astronomen W. J. Foerster zur Vervollkommnung des Öffentlichen Zeitdienstes', *Uhren und Schmuck*, 26 (1989), 54–56.

‘Association for Chronometry’ (*Vereinigung für Chronometrie*) with the objective to encourage the manufacture of chronometers and the production of blank movements in Germany.³⁴ The Admiralty favoured production in Glashütte in Saxony, and already in 1900 the factory of Strasser & Rohde had turned out a number of blank movements which were considered superior to English movements in every respect. Lange & Söhne also started producing complete chronometers. Parallel to these activities – or rather in competition – Dencker planned the production of chronometers on an industrial scale in cooperation with Hamburg shipowners and founded the ‘Hamburg Chronometer Works’ in 1905. He became executive director and developed an ambitious scheme for making chronometers with interchangeable parts,³⁵ entirely different from the system employed in England, which was based on the division of labour in small, highly specialized workshops. In 1907 the company handed in its first six chronometers for examination at the *Seewart*. Of these three were returned after a few days, another turned out to be defective and in two chronometers English parts had been found.³⁶ Dencker’s personnel management left much to be desired and his handling of money was not above suspicion, and therefore the intended annual output of 240 chronometers³⁷ was never realized. The Chronometer Works soon got into financial trouble, and in 1908 Dencker was replaced by his foreman William Meier († 1929), who reverted to manual production. Although Dencker, who bristled with anger, derisively spoke of ‘backwardly oriented house work’ (*rückständige Hausarbeit*), Meier turned

the company into a profitable business. Dencker’s failure is closely linked with his problematic personality. His idea of an industrial production of chronometers was essentially correct and feasible. This was proven later by Hamilton in America – albeit in wartime conditions – which turned out precision timepieces in mass production.³⁸ And the Hamburg Chronometer Works, taken over by Herbert Wempe in 1938 and shortly afterwards subordinated to military authorities, produced the ‘Unified Chronometer’ (*Einheitschronometer*) in cooperation with Lange & Söhne in large quantities to satisfy increased wartime demands.³⁹

In 1910/11 98 per cent of the chronometers participating in the annual trial in Hamburg were of German origin, and in the following year foreign chronometers were generally excluded from trials. The target of creating the ‘German chronometer’ had been met at last. But this was a narrowly defined success limited to the official trials, and the import of English chronometers continued. Participating in the official trials was respectable and printing ‘Purveyor to the Imperial Admiralty’ on the letterhead enhanced reputation considerably. However, producing chronometers in Germany was not a profitable business, and commercial contacts of German makers with England lasted until the eve of the First World War.

Notwithstanding all zealous efforts to promote independent manufacture there was a general and enduring problem both in England and Germany: the low demand for chronometers. Its introduction induced rapid increase in production in England, but the market was quickly saturated.

34. Klaus-Harro Tiemann, ‘Zur Gründungsgeschichte der ‘Vereinigung für Chronometrie’ um 1900’, *Dresdner Beiträge zur Geschichte der Technikwissenschaften*, H. 13 (1987), 42–52; the same in: *Uhren und Schmuck*, 25 (1988), 184–187, 189.

35. Berlin, Geh. Staatsarchiv PK: I. HA Rep. 76, Vc Sekt. 1 Tit. XI Teil 2 Nr. 16, Bd. 1, fol. 311r–318r; Ferdinand Dencker, ‘Das Marinechronometer’, *Hansa*, 38 (1901), pp. 497–500, 510–512, 522–524, 534–537; ‘Das Deutsche Marinechronometer’, *Hansa*, 45 (1908), pp. 687–690.

36. Freiburg/Br., Federal Archive/Military Archive: RM 3/10781, fol. 127r; 205r/v.

37. According to a description of the factory in *Hamburger Nachrichten*, 115 (1906), nr. 887, Dec. 18th.

38. Marvin E. Whitney, *The Ship’s Chronometer*. (Cincinnati, American Watchmakers Institute Press: 1991²), pp. 3–4, 51.

39. See Johannes Altmeppen and Herbert Dittrich, *Das Deutsche Einheits-Chronometer* (Königswinter, Heel: 2012).

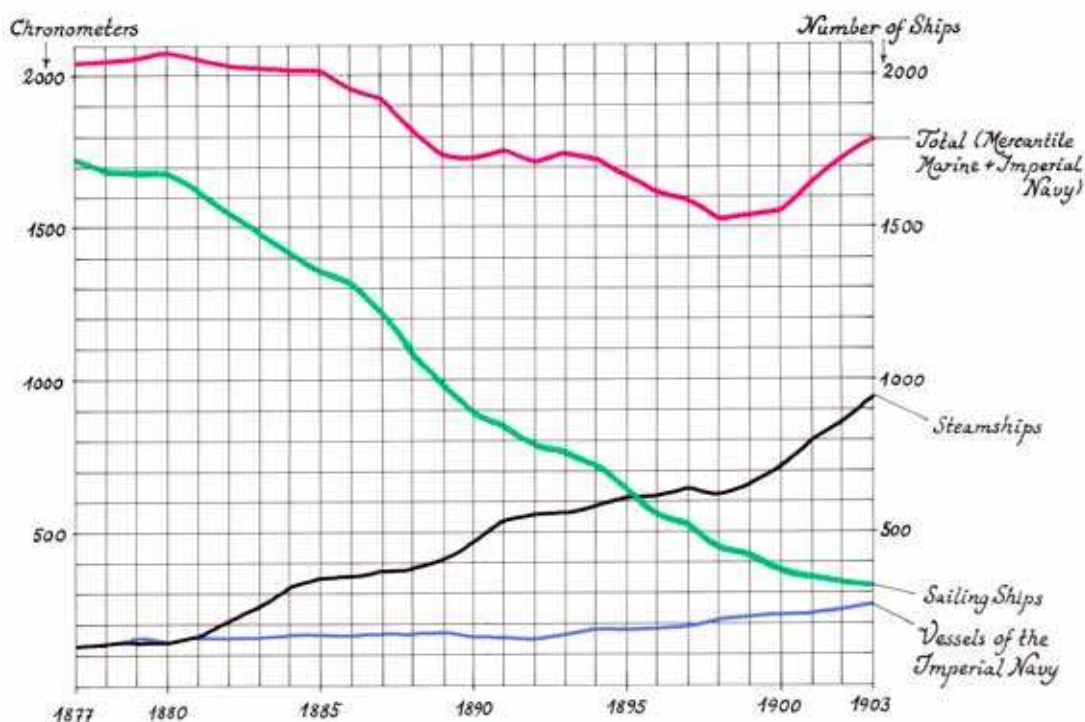


Fig. 11. The presence of chronometers on various types of German ships from 1877 to 1903. Drawing by the author based on data in Knipping, 'Statistik der Schiffschronometer ...' (1904), 232, 234-5.

About 1820 the chronometer was fully developed, and there was no need for replacement by newer and better models. Moreover, these timepieces proved to be extremely durable and could be used on several ship generations, provided they received regular and careful maintenance. A famous specimen, formerly belonging to Nelson's *Victory*, had been purchased by the Admiralty in 1796 and was not discharged until 1907.⁴⁰

Already in 1830 the Royal Navy had acquired more chronometers than were actually needed. The annual demand for replacements amounted to only five or six per year.⁴¹ Even in the phase of reorganization and fierce competition with Germany before 1914 only twenty-four chronometers per year were purchased on average: 'A handful of dreadnoughts, clearly, did not generate a large new demand for chronometers', as Alun Davies aptly put it in his PhD thesis.⁴²

The demand for chronometers by the British merchant marine was also rather moderate. Although tonnage quadrupled between 1840 and 1910, the number of vessels decreased, as their average size increased considerably. Most merchant vessels were utilized in coastal traffic for which chronometers were not needed, and ships for trans-oceanic routes were not so numerous that a significant surge in demand could have been triggered.⁴³ Older chronometers often survived several generations of ships. Therefore both Kullberg and Mercer were looking eagerly for marketing opportunities in Europe and overseas.

The introduction of steamships in the course of the nineteenth century induced profound structural changes in shipping: large numbers of comparatively small sailing ships were replaced by few steamers. In 1877, 1753 German sailing ships were

40. Davies, *The Rise and Decline*, p. 189.

41. Davies, *The Rise and Decline*, p. 190.

42. Davies, *The Rise and Decline*, p. 191.

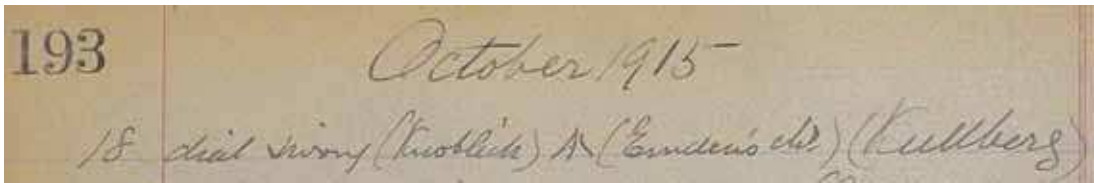


Fig. 12. Entry concerning a chronometer from the *Emden* in the Day Book of Thomas Mercer, which contains commissions for the engravings of chronometer dials and ivory labels between 1905 and 1917. (Liverpool, National Museums/World Museum)

equipped with at least one chronometer, but their number diminished to 463 in 1903. In contrast, the number of steamers rose from 166 in 1877 to 950 in 1903. Registered tonnage of the German mercantile marine increased more than sevenfold between 1850 and 1910 and ever bigger ships were launched, but their total number went down.⁴⁴ In 1877, the ships of the German mercantile marine were equipped with 1919 chronometers, and the Imperial Navy possessed 126 chronometers. In 1903 the ratio was 1413 against 242.⁴⁵

It was common to carry three chronometers on each ship of the Imperial Navy, but notwithstanding all efforts for maritime arming, demand from the Admiralty alone was insufficient for the survival of Germany's chronometer makers. Because chronometers were valuable and expensive instruments they usually received careful refurbishment and servicing. Their very durability reduced the demand for replacements. Thus the production of chronometers in Germany was much hampered and limited by the nature of demand, inherent supply problems and partly encouraged, but also constrained, by imposed governmental regulation.

The story of the 'German chronometer' may be concluded with a sort of epilogue. The raids and fate of the German cruiser *Emden*, a modern corsair of the First World War, belong to the most famous and well-known stories of maritime history. In November 1914, lieutenant commander Karl



Fig. 13. The wreck of the German cruiser *Emden*. (Melbourne, State Library of Victoria)

von Müller (1873–1923) planned to destroy the strategically important cable station situated on Direction Island (Cocos Islands) in the Indian Ocean, and with this surprise attack the ship's last hour came. It was caught off guard by the superior Australian cruiser HMAS *Sydney* and immediately attacked. *Emden* was soon heavily damaged, and Commander von Müller ran the ship aground on North Keeling Island to rescue his remaining crew (Fig. 13). Despite heavy shelling some equipment had obviously remained intact, and one of *Emden*'s chronometers was transferred to England. There it received a new dial with the 'Broad Arrow' (Fig. 12), which characterized it as British Government property.

In its very first battle, the young Royal Australian Navy had conquered a 'German' chronometer for Albion, which most likely had been made in England. Sometimes the muse of historiography displays a sense of subtle irony.

43. Davies, *The Rise and Decline*, pp. 192–3.

44. Walther Vogel, *Die deutsche Handelsmarine im 19. Jahrhundert*, Meereskunde: Sammlung volkstümlicher Vorträge zum Verständnis der nationalen Bedeutung von Meer und Seewesen, H. 85 (Berlin, Ernst Siegfried Mittler & Sohn: 1914), p. 6.

45. Erwin Knipping, 'Statistik der Schiffschronometer der deutschen Kriegs- und Handelsmarine von 1877 bis 1903', *Annalen der Hydrographie und Maritimen Meteorologie: Zeitschrift für Seefahrts- und Meereskunde*, 32 (1904), 231–238.